

Module 3

CHY1009 – Chemistry and Environmental Studies

Electrochemistry and Chemical Sensors

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- Electrochemical cell
 - Galvanic cell
 - **Electrolytic cell**
- Cell representation
- Cell potential
- SHE
- Electrochemical series
- Standard cell EMF

$$E_{cell}^0 = E_{cathode}^0 - E_{Anode}^0$$

$$\Delta G^\circ = -nFE_{cell}^\circ$$

$$\Delta G^\circ = -RT \ln K$$

$$-nFE_{cell}^\circ = -RT \ln K$$

$$E_{cell}^\circ = \frac{0.0592 \text{ V}}{n} \log K \quad @RT$$

$$E_{cell} = E_{cell}^\circ - \frac{0.0592}{n} \log_{10} Q$$

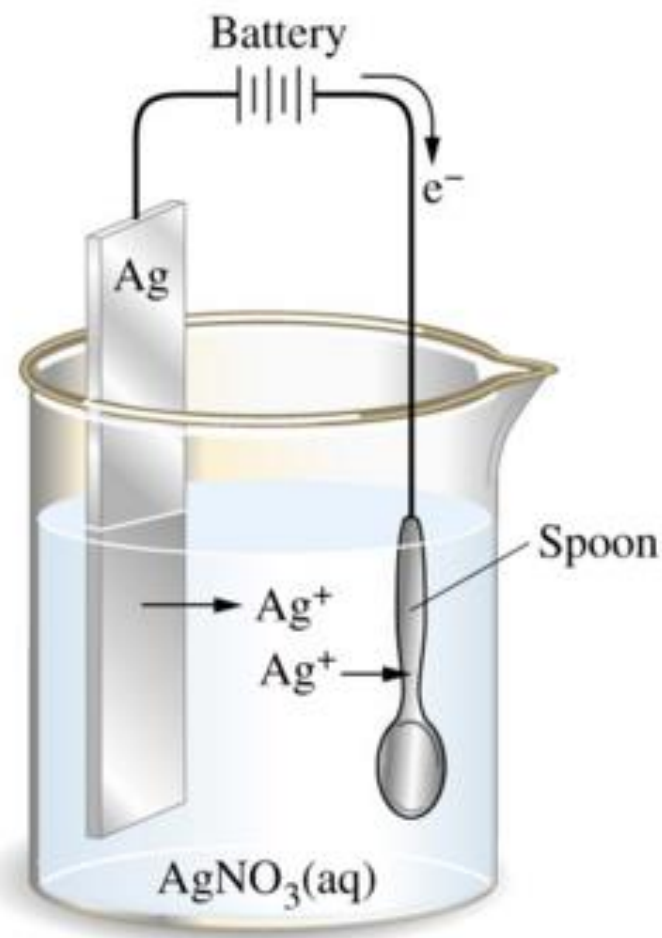
Nernst equation

Electrolytic cell

CHEMICAL ENERGY

ELECTRICAL ENERGY

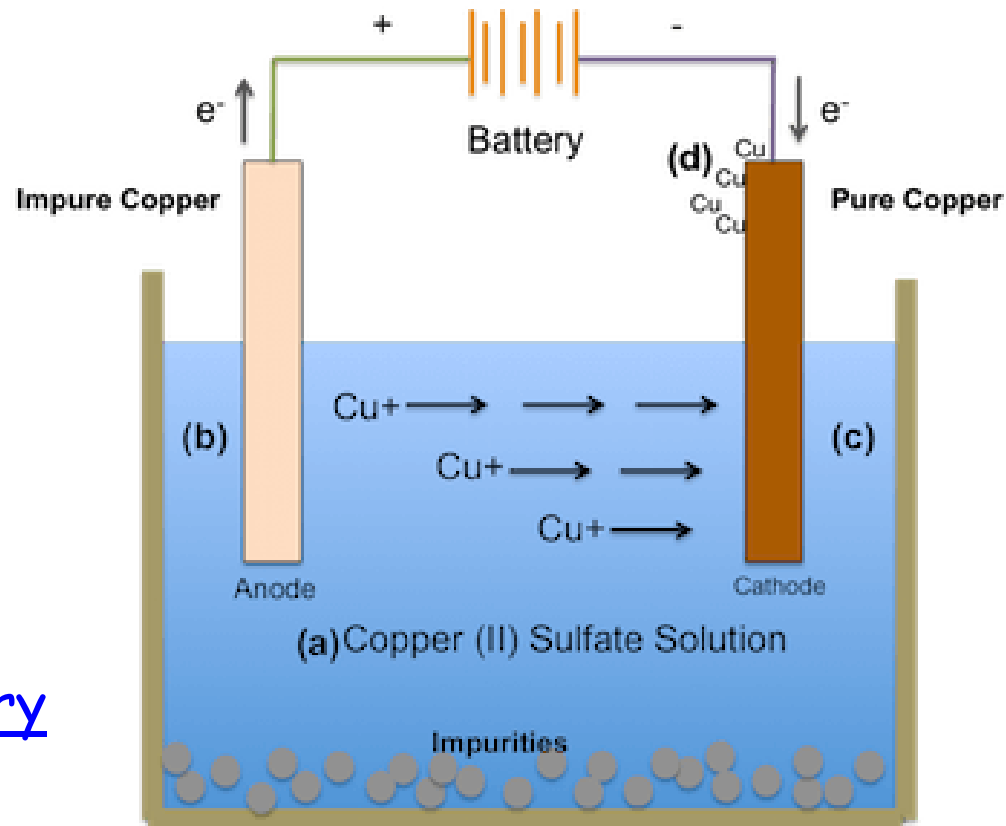
- An **electrolytic cell** uses electrical energy to drive a **non-spontaneous** reaction ($\Delta G > 0$).
- In the cell reaction, electrical energy from an external power supply converts lower energy reactants into higher energy products.
- Thus, *the surroundings do work on the system*.
- **Electroplating** and **recovering metals** from ores involve electrolytic cells.



Electrolytic cell

- Passage of electricity through an electrolyte
- The **cations** are moving to **cathode** and **reduced**
- The **anions** are moving to **anode**, and **oxidised**

The positive ions move to the electrodes connected with the negative terminal of the battery where these positive ions take electrons from it, becoming a pure metal atom and getting deposited on the electrode



Faraday's Laws of Electrolysis

1st Law:

"the mass of a substance deposited or liberated at any electrode is directly proportional to the amount of charge passed"

Or

The amount of chemical change produced by current at an electrode-electrolyte boundary is proportional to the quantity of electricity used

$$m \propto [\text{quantity of electricity}]$$

$$m \propto q$$

$$m = z q$$

m = mass of substance deposited

q = amount of charge passes

z = Electrochemical equivalent

When $Q = 1$, then $z = m$

$$m = z q$$

$$\text{Current (i)} = \frac{\text{charge (q)}}{\text{time (t)}}$$

$$\text{Charge (q)} = i \times t$$

$$m = z \times i \times t$$

$$m = z i t$$

Calculation of electrochemical equivalent (**Z**)

$$Z = \frac{E}{F}$$

E – Equivalent weight of Substance

F – Faraday's Constant

Faraday's Constant (F): Charge on one mole of electron
= charge of one electron $\times N_A = 96,500 \text{ C}$

2nd Law

“the mass of a substance deposited or liberated at any electrode on passing a certain amount of charge is directly proportional to its chemical equivalent weight”

Or

the amounts of chemical changes produced by the given quantity of electricity in different substances are proportional to their equivalent weights

Or

For the same quantity of electricity or charge passes through different electrolytes, the mass of deposited chemical is directly proportional to its atomic weight and inversely proportional to its valency.

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

W is the mass of the substance in grams

E is its chemical equivalent weight in gms per equivalent

Equivalent weight:

- The weight of an element that is liberated in an electrolysis (chemical reaction caused by an electric current) by the passage of 96,500 coulombs of electricity.

$$\text{chemical equivalent} = \frac{\text{Atomic weight}}{\text{Valency}}$$

Faraday's Constant (F)

- The quantity of electricity that will cause a chemical change of one equivalent weight unit has been designated a faraday (F).
- It is equivalent to 9.6485309×10^4 coulombs of electricity

$$\begin{aligned} F &= N_A \times e = 6.023 \times 10^{23} \text{ mol}^{-1} \times 1.602 \times 10^{-19} \text{ C} \\ &= 96485 \text{ C mol}^{-1} \end{aligned}$$

Example:

In the electrolysis of fused magnesium chloride (MgCl_2) one faraday of electricity will deposit $24.312/2$ grams of magnesium at the negative electrode and liberate 35.453 grams of chlorine at the positive electrode.

Summary:

$$m = z q$$

$$m = z i t$$

$$Z = \frac{E}{F}$$

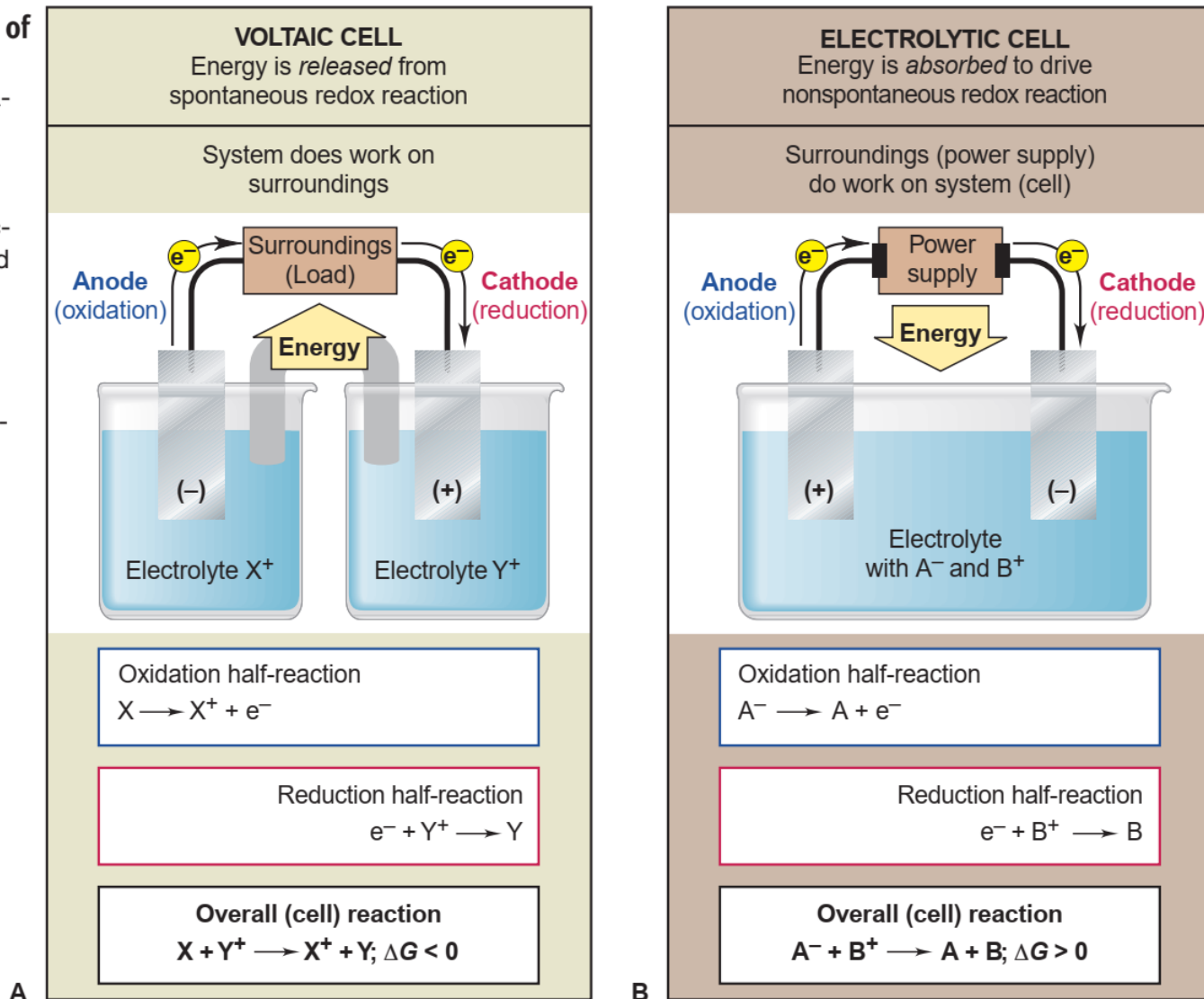
$$E = \frac{\textit{Atomic weight}}{\textit{Valancy}} = \frac{\textit{Atomic weight}}{\textit{No of electron}} = \frac{MW}{n}$$

$$Z = \frac{\textit{Molecular weight}}{nF}$$

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

Characteristics of Voltaic and Electrolytic Cells

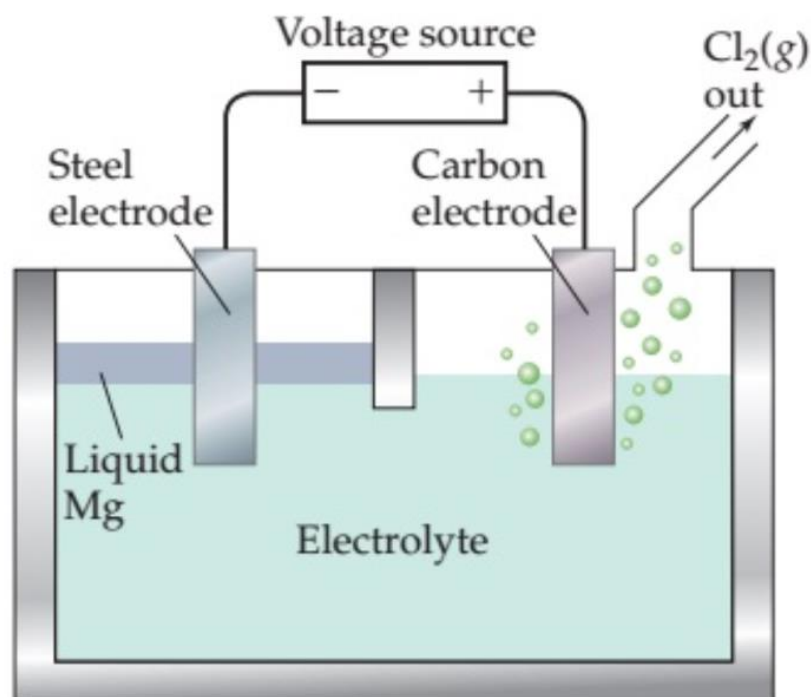
Figure 21.3 General characteristics of voltaic and electrolytic cells. A voltaic cell (A) generates energy from a spontaneous reaction ($\Delta G < 0$), whereas an electrolytic cell (B) requires energy to drive a nonspontaneous reaction ($\Delta G > 0$). In both types of cell, two electrodes dip into electrolyte solutions, and an external circuit provides the means for electrons to flow between them. Most important, notice that oxidation takes place at the anode and reduction takes place at the cathode, but the relative electrode charges are opposite in the two cells.



Problem

Calculation of amount of metals deposited

Magnesium, the element, is produced commercially by electrolysis from a molten salt (the "electrolyte") using a cell similar to the one shown below.



QN

**60000 coulombs (C) of charge was passed in the circuit.
Calculate the amount of magnesium deposited at the cathode during the electrolysis?**

Atomic weight of Mg = 24.3

Solution

Step 1.

Find the **number of electrons** involved in the redox reaction.
This can be done by writing the **cell reaction** and the half cell reactions



Therefore, the reaction is two electron process or $n = 2$

Step 2.

$$\text{mass deposited (m)} = z q = \frac{\text{Atomic weight}}{nF} \times Q$$

Given,

Atomic weight of Mg = 24.3

Charge (Q) = 60,000 C

$$\begin{aligned} \text{Mass deposited (m)} &= \frac{24.3}{2 \times 96500} \times 60000 \\ &= 7.56 \text{ g} \end{aligned}$$

Problem

The electrolysis of molten MgCl_2 is the last step in the isolation of magnesium metal from seawater. How many amps are needed to produce 35.6 g of Mg in 2.50 hours?

Atomic weight of Mg = 24.3

Solution

Given,

Atomic weight of Mg = 24.3

Mass deposited (m) = 35.6 g

Time (t) = 2.5 hr = $2.5 \times 60 \times 60 = 9000$ sec

Current (i) = ?

$$m = z i t = \frac{\text{Atomic weight}}{nF} \times (i) \times t$$

$$\text{Mass deposited} = \frac{24.3}{2 \times 96500} \times (i) \times 9000 = 35.6$$

$$\text{Current (i)} = 31.4 \text{ A}$$

Problem

A galvanic cell consisting of standard concentrations: $\text{Cu}|\text{Cu}^{2+}||\text{Ag}^{+}|\text{Ag}$, is connected to a small light bulb for 40.0 minutes. During that time the average current drawn by the bulb is 0.12 A. How many grams of copper dissolve from the anode during this time?

Atomic weight of Cu = 63.55

Ans. = 0.095 g

Problem

A lead storage battery like one used to start your car contains lead and lead(IV) oxide electrodes in a sulfuric acid electrolyte. The electrode reactions are:



A typical automobile battery might be rated at "100 ampere-hours". This means it can deliver a current of 1.0 A for 100 hours.

How many grams of Pb are oxidized for the full rating?

Atomic weight of Pb = 207.2

Number of coulombs of charge passed (Q) = Current in amperes (A) x time in seconds (s)

$$Q = It \quad (\text{unit is A s})$$

$$Q = 1 \times 100 \times 60 \times 60 = 360000 \text{ C}$$

$$= \frac{207.2 \text{ g}}{193000 \text{ C}} \times 360000 \text{ C} = 386.5 \text{ g}$$

Problem

Calculations for the Gaseous Products

Calculate the volume of hydrogen (H_2) produced (measured at room temperature and pressure -RTP) during the electrolysis of dilute sulphuric acid the current flow 1.0 amp for 15 minutes.

Step 1.

We know that, $F = 96500 \text{ C mol}^{-1}$

The molar volume of a gas at rtp = $24 \text{ dm}^3 \text{ mol}^{-1}$.

Step 2.

The number of coulombs of charge flowed (Q) = $i \times t$

$$Q = 1.0 \times 15 \times 60 = 900 \text{ C}$$

Step 3.

Find the number of electron involved in formation of H_2 ?



Step 4.

$$\text{Volume of } \text{H}_2 \text{ produce} = \frac{\text{molar volume}}{nF} \times Q = \frac{24}{2 \times 96500} \times 900 = 0.11 \text{ dm}^3$$

Problem

A voltaic cell consists of a metal A/A^+ electrode and a metal B/B^+ electrode, with the A/A^+ electrode negative. The initial $[A^+]/[B^+]$ is such that $E_{\text{cell}} > E_{\text{cell}}^{\circ}$.

- (a) How do $[A^+]$ and $[B^+]$ change as the cell operates?
- (b) How does E_{cell} change as the cell operates?
- (c) What is $[A^+]/[B^+]$ when $E_{\text{cell}} = E_{\text{cell}}^{\circ}$? Explain.
- (d) Is it possible for E_{cell} to be less than E_{cell}° ? Explain.

Solution

a) $[A^+]$ increases and $[B^+]$ decreases.

b) E_{cell} decreases.

c) $E_{\text{cell}} = E_{\text{cell}}^0 - (RT/nF)\ln([A^+]/[B^+]); E_{\text{cell}} = E_{\text{cell}}^0$
when $(RT/nF)\ln([A^+]/[B^+]) = 0$, This occurs
when $\ln([A^+]/[B^+]) = 0$, that is, $[A^+]$ equals
 $[B^+]$.

d) Yes, when $[A^+] > [B^+]$.